Week 7 Assignment

Delta Lake MERGE Operation using Apache Spark (PySpark)

Objective

The objective of this assignment is to understand and implement Delta Lake MERGE operations using Apache Spark (PySpark). The assignment covers loading and cleaning data, creating Delta Tables, generating incremental datasets, performing MERGE operations to update existing records and insert new records, validating the results, and understanding how Delta Lake enables efficient, reliable, and ACID-compliant data processing for incremental data ingestion.

Step 1: Install Required Libraries

Description

Installed the required libraries for Apache Spark and Delta Lake in the Google Colab environment.

Libraries Installed:

- PySpark
- Delta Spark

Screenshot

```
[1]
10s
!pip install pyspark delta-spark

Requirement already satisfied: pyspark in /usr/local/lib/python3.12/dist-packages (4.0.3)
Collecting delta-spark
  Downloading delta_spark-4.3.1-py3-none-any.whl.metadata (2.2 kB)
Requirement already satisfied: py4j<0.10.9.10,>=0.10.9.7 in /usr/local/lib/python3.12/dist-packages (from pyspark) (0.10.9.9)
Collecting importlib_metadata<=8.7.1,>=1.0.0 (from delta-spark)
  Downloading importlib_metadata-8.7.1-py3-none-any.whl.metadata (4.7 kB)
Requirement already satisfied: zipp>=3.20 in /usr/local/lib/python3.12/dist-packages (from importlib_metadata<=8.7.1,>=1.0.0->delta-spark) (4
Downloading delta_spark-4.3.1-py3-none-any.whl (53 kB)
   ---------------------------------------- 53.9/53.9 kB 2.1 MB/s eta 0:00:00
Downloading importlib_metadata-8.7.1-py3-none-any.whl (27 kB)
Installing collected packages: importlib_metadata, delta-spark
  Attempting uninstall: importlib_metadata
    Found existing installation: importlib_metadata 9.0.0
    Uninstalling importlib_metadata-9.0.0:
      Successfully uninstalled importlib_metadata-9.0.0
Successfully installed delta-spark-4.3.1 importlib_metadata-8.7.1
```

Step 2: Import Libraries and Configure Spark Session

Description

Imported the required PySpark and Delta Lake libraries. Configured a Spark Session with Delta Lake extensions to enable Delta table creation and MERGE operations.

Libraries Imported

- pyspark
- delta
- SparkSession

Configuration

- Enabled Delta Lake SQL Extension.
- Configured Delta Catalog.
- Created Spark Session with Delta Lake support.

Screenshot

```python
import pyspark
from delta import configure_spark_with_delta_pip
from pyspark.sql import SparkSession
```

```python
builder = SparkSession.builder \
    .appName("Week7_DeltaLake_Assignment") \
    .config("spark.sql.extensions", "io.delta.sql.DeltaSparkSessionExtension") \
    .config("spark.sql.catalog.spark_catalog", "org.apache.spark.sql.delta.catalog.DeltaCatalog")

spark = configure_spark_with_delta_pip(builder).getOrCreate()

print("Spark Session with Delta Lake Created Successfully!")
```

```
Spark Session with Delta Lake Created Successfully!
```

Step 3: Load Master Dataset

Description

The **Sample - Superstore.csv** dataset was uploaded into Google Colab and loaded into a Spark DataFrame using spark.read.csv(). The first row was treated as the header, and Spark automatically inferred the schema using inferSchema=True.

Configuration

- File Format: CSV
- Header: Enabled
- Schema Inference: Enabled

Dataset Information

- Dataset Name: Sample - Superstore
- Records: **9994**
- Columns: **21**

Verification

The dataset was verified by:

- Displaying the first five records using show(5).
- Printing the schema using printSchema().
- Counting the total number of records and columns.

Screenshots

```
from google.colab import files

uploaded = files.upload()
```

```
Choose Files  Sample - Superstore.csv
Sample - Superstore.csv(text/csv) - 2287806 bytes, last modified: 8/1/2026 - 100% done
Saving Sample - Superstore.csv to Sample - Superstore.csv
```

```
df = spark.read.csv(
    "Sample - Superstore.csv",
    header=True,
    inferSchema=True
)

print("Dataset Loaded Successfully!")
```

```
Dataset Loaded Successfully!
```

```
df.show(5)
```

```
+------+--------------+----------+----------+--------------+-----------+----------------+---------+-------------+---------------+----------+--
|Row ID|      Order ID|Order Date| Ship Date|     Ship Mode|Customer ID|   Customer Name|  Segment|      Country|           City|     State|Po
+------+--------------+----------+----------+--------------+-----------+----------------+---------+-------------+---------------+----------+--
|     1|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|   CG-12520|     Claire Gute| Consumer|United States|      Henderson|  Kentucky|
|     2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|   CG-12520|     Claire Gute| Consumer|United States|      Henderson|  Kentucky|
|     3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|   DV-13045|Darrin Van Huff|Corporate|United States|    Los Angeles|California|
|     4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|   SO-20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|   Florida|
|     5|US-2015-108966|10/11/2015|10/18/2015|Standard Class|   SO-20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|   Florida|
+------+--------------+----------+----------+--------------+-----------+----------------+---------+-------------+---------------+----------+--
only showing top 5 rows
```

```
df.printSchema()
```

```
root
 |-- Row ID: integer (nullable = true)
 |-- Order ID: string (nullable = true)
 |-- Order Date: string (nullable = true)
 |-- Ship Date: string (nullable = true)
 |-- Ship Mode: string (nullable = true)
 |-- Customer ID: string (nullable = true)
 |-- Customer Name: string (nullable = true)
 |-- Segment: string (nullable = true)
 |-- Country: string (nullable = true)
 |-- City: string (nullable = true)
 |-- State: string (nullable = true)
 |-- Postal Code: integer (nullable = true)
 |-- Region: string (nullable = true)
 |-- Product ID: string (nullable = true)
 |-- Category: string (nullable = true)
 |-- Sub-Category: string (nullable = true)
 |-- Product Name: string (nullable = true)
 |-- Sales: string (nullable = true)
 |-- Quantity: string (nullable = true)
 |-- Discount: string (nullable = true)
 |-- Profit: double (nullable = true)
```

```
print("Total Records :", df.count())
print("Total Columns :", len(df.columns))
```

```
Total Records : 9994
Total Columns : 21
```

Step 4: Data Cleaning

Description

The dataset was analyzed for data quality before performing Delta Lake operations.

The following checks were performed:

- Checked for null values in all columns.
- Checked for duplicate records.
- Removed duplicates (if any).
- Removed null values (if any).

Observation

- Total Records: **9994**
- Null Values: **0**
- Duplicate Records: **0**

The dataset was already clean and did not require additional preprocessing.

Screenshot

```python
from pyspark.sql.functions import col, count, when

df.select([
    count(when(col(c).isNull(), c)).alias(c)
    for c in df.columns
]).show()
```

```
+------+--------+----------+---------+---------+-----------+-------------+-------+-------+----+-----+-----------+------+----------+--------+-
|Row ID|Order ID|Order Date|Ship Date|Ship Mode|Customer ID|Customer Name|Segment|Country|City|State|Postal Code|Region|Product ID|Category|S
+------+--------+----------+---------+---------+-----------+-------------+-------+-------+----+-----+-----------+------+----------+--------+-
|     0|       0|         0|        0|        0|          0|            0|      0|      0|   0|    0|          0|     0|         0|       0|
+------+--------+----------+---------+---------+-----------+-------------+-------+-------+----+-----+-----------+------+----------+--------+-
```

```python
total_rows = df.count()

duplicate_rows = total_rows - df.dropDuplicates().count()

print("Total Rows :", total_rows)
print("Duplicate Rows :", duplicate_rows)
```

```
Total Rows : 9994
Duplicate Rows : 0
```

Step 5: Create Delta Table

Description

Before creating the Delta Table, the column names were standardized by replacing spaces ( ) and hyphens (-) with underscores (_). This step was necessary because Delta Lake does not allow certain special characters in column names.

After updating the schema, the cleaned DataFrame was stored in **Delta format** using the write.format("delta") method. The Delta Table was then loaded and verified by displaying sample records and checking the total number of rows.

Schema Modification

Original Column	**Updated Column**
Row ID	Row_ID
Order ID	Order_ID
Order Date	Order_Date
Ship Date	Ship_Date
Ship Mode	Ship_Mode
Customer ID	Customer_ID
Customer Name	Customer_Name
Postal Code	Postal_Code
Product ID	Product_ID
Product Name	Product_Name
Sub-Category	Sub_Category

Code Used

- Renamed column names
- Created Delta Table
- Loaded Delta Table
- Displayed sample records

Result

The Delta Table was successfully created from the cleaned dataset and will be used in the next step to perform **MERGE** operations with the incremental dataset.

Screenshots

```
clean_df = clean_df.toDF(
    *[
        c.replace(" ", "_").replace("-", "_")
        for c in clean_df.columns
    ]
)
```

```
clean_df.write \
    .format("delta") \
    .mode("overwrite") \
    .save("delta_superstore")
```

```
from delta.tables import DeltaTable

delta_table = DeltaTable.forPath(
    spark,
    "delta_superstore"
)

print("Delta Table Created Successfully!")
```

```
Delta Table Created Successfully!
```

```
delta_table.toDF().show(5)
```

```
+------+--------------+----------+----------+--------------+-----------+---------------+---------+-------------+---------------+----------+--
|Row_ID|      Order_ID|Order_Date| Ship_Date|     Ship_Mode|Customer_ID|  Customer_Name|  Segment|      Country|           City|     State|Po
+------+--------------+----------+----------+--------------+-----------+---------------+---------+-------------+---------------+----------+--
|     1|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|   CG-12520|    Claire Gute| Consumer|United States|      Henderson|  Kentucky|
|     2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|   CG-12520|    Claire Gute| Consumer|United States|      Henderson|  Kentucky|
|     3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|   DV-13045|Darrin Van Huff|Corporate|United States|    Los Angeles|California|
|     4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|   SO-20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|   Florida|
|     5|US-2015-108966|10/11/2015|10/18/2015|Standard Class|   SO-20335| Sean O'Donnell| Consumer|United States|Fort Lauderdale|   Florida|
+------+--------------+----------+----------+--------------+-----------+---------------+---------+-------------+---------------+----------+--
only showing top 5 rows
```

```
print("Rows in Delta Table :", delta_table.toDF().count())
```

```
Rows in Delta Table : 9994
```

Step 6: Create Incremental Dataset

Description

An incremental dataset containing four records was created.

The dataset includes:

- **2 existing records** (Row_ID 1 and 2) with modified values to demonstrate **UPDATE** operations.
- **2 new records** (Row_ID 9995 and 9996) to demonstrate **INSERT** operations.

The Row_ID column was used as the merge key because it uniquely identifies each record in the dataset.

Incremental Data Summary

Operation	Number of Records
Update	2
Insert	2
Total	4

Result

The incremental dataset was successfully created and is ready to be merged with the Delta Table.

Screenshots

```
clean_df.select(
    "Row_ID",
    "Order_ID",
    "Customer_Name",
    "Sales",
    "Profit"
).show(10, truncate=False)
```

```
+------+--------------+--------------+--------+--------+
|Row_ID|Order_ID      |Customer_Name |Sales   |Profit  |
+------+--------------+--------------+--------+--------+
|1     |CA-2016-152156|Claire Gute   |261.96  |41.9136 |
|2     |CA-2016-152156|Claire Gute   |731.94  |219.582 |
|3     |CA-2016-138688|Darrin Van Huff|14.62  |6.8714  |
|4     |US-2015-108966|Sean O'Donnell|957.5775|-383.031|
|5     |US-2015-108966|Sean O'Donnell|22.368  |2.5164  |
|6     |CA-2014-115812|Brosina Hoffman|48.86  |14.1694 |
|7     |CA-2014-115812|Brosina Hoffman|7.28   |1.9656  |
|8     |CA-2014-115812|Brosina Hoffman|907.152|90.7152 |
|9     |CA-2014-115812|Brosina Hoffman|18.504 |5.7825  |
|10    |CA-2014-115812|Brosina Hoffman|114.9  |34.47   |
+------+--------------+--------------+--------+--------+
only showing top 10 rows
```

```
incremental_df.show(truncate=False)
```

```
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+---------+--------+-----------+-------+-----------
|Row_ID|Order_ID      |Order_Date|Ship_Date |Ship_Mode     |Customer_ID|Customer_Name|Segment  |Country      |City     |State   |Postal_Code|Region |Product_ID
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+---------+--------+-----------+-------+-----------
|1     |CA-2016-152156|11/8/2016 |11/11/2016|Second Class  |CG-12520   |Claire Gute  |Consumer |United States|Henderson|Kentucky|42420      |East   |FUR-BO-100
|2     |CA-2016-152156|11/8/2016 |11/11/2016|Second Class  |CG-12520   |Claire Gute  |Consumer |United States|Henderson|Kentucky|42420      |East   |FUR-CH-100
|9995  |CA-2024-999995|01/08/2024|05/08/2024|Standard Class|HS-50001   |Harshit Goel |Consumer |United States|New York |New York|10001      |East   |TEC-100001
|9996  |CA-2024-999996|02/08/2024|06/08/2024|First Class   |HS-50002   |Aman Sharma  |Corporate|United States|Chicago  |Illinois|60601      |Central|OFF-100002
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+---------+--------+-----------+-------+-----------
```

Step 7: Perform MERGE Operation

Description

The MERGE operation was performed using **Row_ID** as the unique merge key.

The following actions were executed:

- Updated existing records (Row_ID 1 and 2).
- Inserted new records (Row_ID 9995 and 9996).

The operation used:

- **WHEN MATCHED UPDATE**
- **WHEN NOT MATCHED INSERT**

Merge Condition

target.Row_ID = source.Row_ID

Updated Columns

- Sales
- Profit
- Region

Result

The Delta Table was successfully synchronized with the incremental dataset.

- Existing records were updated.
- New records were inserted.

Screenshots

```
from delta.tables import DeltaTable

delta_table.alias("target").merge(
    incremental_df.alias("source"),
    "target.Row_ID = source.Row_ID"
).whenMatchedUpdate(
    set={
        "Sales": "source.Sales",
        "Profit": "source.Profit",
        "Region": "source.Region"
    }
).whenNotMatchedInsertAll().execute()

print("MERGE Operation Completed Successfully!")
```

```
MERGE Operation Completed Successfully!
```

```
delta_table.toDF() \
    .filter("Row_ID IN (1,2)") \
    .show(truncate=False)
```

```
+------+--------------+----------+----------+------------+-----------+-------------+--------+-------------+---------+--------+-----------+------+---------------
|Row_ID|Order_ID      |Order_Date|Ship_Date |Ship_Mode   |Customer_ID|Customer_Name|Segment |Country      |City     |State   |Postal_Code|Region|Product_ID
+------+--------------+----------+----------+------------+-----------+-------------+--------+-------------+---------+--------+-----------+------+---------------
|2     |CA-2016-152156|11/8/2016 |11/11/2016|Second Class|CG-12520   |Claire Gute  |Consumer|United States|Henderson|Kentucky|42420      |East  |FUR-CH-1000045
|1     |CA-2016-152156|11/8/2016 |11/11/2016|Second Class|CG-12520   |Claire Gute  |Consumer|United States|Henderson|Kentucky|42420      |East  |FUR-BO-1000179
+------+--------------+----------+----------+------------+-----------+-------------+--------+-------------+---------+--------+-----------+------+---------------
```

```
delta_table.toDF() \
    .filter("Row_ID >= 9995") \
    .show(truncate=False)
```

```
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+-----------+-
|Row_ID|Order_ID      |Order_Date|Ship_Date |Ship_Mode     |Customer_ID|Customer_Name|Segment  |Country      |City    |State   |Postal_Code|Region |Product_ID|(
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+-----------+-
|9995  |CA-2024-999995|01/08/2024|05/08/2024|Standard Class|HS-50001   |Harshit Goel |Consumer |United States|New York|New York|10001      |East   |TEC-100001|T
|9996  |CA-2024-999996|02/08/2024|06/08/2024|First Class   |HS-50002   |Aman Sharma  |Corporate|United States|Chicago |Illinois|60601      |Central|OFF-100002|(
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+-----------+-
```

```
print("Total Rows After MERGE :", delta_table.toDF().count())
```

```
Total Rows After MERGE : 9996
```

Step 8: Validate MERGE Operation

Objective

Validate that the Delta Lake MERGE operation updated existing records and inserted new records correctly.

Validation Performed

The following validations were carried out:

- Compared the total number of rows before and after the MERGE operation.
- Verified that the records with **Row_ID 1 and 2** were updated successfully.
- Verified that the records with **Row_ID 9995 and 9996** were inserted successfully.
- Displayed the final Delta Table for confirmation.

Validation Result

Validation	Status
Existing Records Updated	Success
New Records Inserted	Success
Final Row Count Verified	Success

Final Observation

The Delta Lake MERGE operation successfully synchronized the master dataset with the incremental dataset. Existing records were updated, new records were inserted, and the integrity of the dataset was maintained.

Screenshots

```
before_merge = clean_df.count()
after_merge = delta_table.toDF().count()

print("Rows Before MERGE :", before_merge)
print("Rows After MERGE  :", after_merge)
```

```
Rows Before MERGE : 9994
Rows After MERGE  : 9996
```

```
delta_table.toDF() \
.select("Row_ID","Sales","Profit","Region") \
.filter("Row_ID IN (1,2)") \
.show(truncate=False)
```

```
+------+-----+------+------+
|Row_ID|Sales|Profit|Region|
+------+-----+------+------+
|2     |900.0|250.0 |East  |
|1     |500.0|120.0 |East  |
+------+-----+------+------+
```

```
delta_table.toDF() \
.filter("Row_ID IN (9995,9996)") \
.show(truncate=False)
```

```
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+----------+-
|Row_ID|Order_ID      |Order_Date|Ship_Date |Ship_Mode     |Customer_ID|Customer_Name|Segment  |Country      |City    |State   |Postal_Code|Region |Product_ID|
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+----------+-
|9995  |CA-2024-999995|01/08/2024|05/08/2024|Standard Class|HS-50001   |Harshit Goel |Consumer |United States|New York|New York|10001      |East   |TEC-100001|
|9996  |CA-2024-999996|02/08/2024|06/08/2024|First Class   |HS-50002   |Aman Sharma  |Corporate|United States|Chicago |Illinois|60601      |Central|OFF-100002|
+------+--------------+----------+----------+--------------+-----------+-------------+---------+-------------+--------+--------+-----------+-------+----------+-
```

```
delta_table.toDF().show(10)
```

```
+------+--------------+----------+----------+--------------+-----------+----------------+-----------+-------------+---------------+--------------+-----------+--
|Row_ID|      Order_ID|Order_Date| Ship_Date|     Ship_Mode|Customer_ID|   Customer_Name|    Segment|      Country|           City|         State|Postal_Code| Re
+------+--------------+----------+----------+--------------+-----------+----------------+-----------+-------------+---------------+--------------+-----------+--
|     2|CA-2016-152156| 11/8/2016|11/11/2016|  Second Class|   CG-12520|     Claire Gute|   Consumer|United States|      Henderson|      Kentucky|      42420|
|     3|CA-2016-138688| 6/12/2016| 6/16/2016|  Second Class|   DV-13045|Darrin Van Huff|  Corporate|United States|    Los Angeles|    California|      90036|
|     4|US-2015-108966|10/11/2015|10/18/2015|Standard Class|   SO-20335| Sean O'Donnell|   Consumer|United States|Fort Lauderdale|       Florida|      33311|
|     8|CA-2014-115812|  6/9/2014| 6/14/2014|Standard Class|   BH-11710|Brosina Hoffman|   Consumer|United States|    Los Angeles|    California|      90032|
|    11|CA-2014-115812|  6/9/2014| 6/14/2014|Standard Class|   BH-11710|Brosina Hoffman|   Consumer|United States|    Los Angeles|    California|      90032|
|    12|CA-2014-115812|  6/9/2014| 6/14/2014|Standard Class|   BH-11710|Brosina Hoffman|   Consumer|United States|    Los Angeles|    California|      90032|
|    13|CA-2017-114412| 4/15/2017| 4/20/2017|Standard Class|   AA-10480|    Andrew Allen|   Consumer|United States|        Concord|North Carolina|      28027|
|    14|CA-2016-161389| 12/5/2016|12/10/2016|Standard Class|   IM-15070|    Irene Maddox|   Consumer|United States|        Seattle|    Washington|      98103|
|    15|US-2015-118983|11/22/2015|11/26/2015|Standard Class|   HP-14815|   Harold Pawlan|Home Office|United States|     Fort Worth|         Texas|      76106|Cer
|    16|US-2015-118983|11/22/2015|11/26/2015|Standard Class|   HP-14815|   Harold Pawlan|Home Office|United States|     Fort Worth|         Texas|      76106|Cer
+------+--------------+----------+----------+--------------+-----------+----------------+-----------+-------------+---------------+--------------+-----------+--
only showing top 10 rows
```

Conclusion

This assignment provided practical experience in implementing **Delta Lake MERGE operations** using **Apache Spark (PySpark)**. The master dataset was loaded, cleaned, and converted into a Delta Table. An incremental dataset was then created to simulate real-world updates and new records. Using the **MERGE** operation, existing records were successfully updated while new records were inserted into the Delta Table. Finally, the results were validated by verifying the updated records, newly inserted records, and the final row count. This assignment strengthened the understanding of Delta Lake, incremental data processing, ACID transactions, and efficient data management techniques used in modern Data Engineering workflows.